Energy-Efficient 200 Gbit/s Transceivers for Optical Access Networks

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Contents

Chapter 1

Access Networks and Coherent PON

1.1 Optical Access Networks

The access network is the final link between an operator's central exchange and the end user, often called the *last mile*. Exponentially growing bandwidth demands from streaming, cloud services, and 5G backhaul are placing severe pressure on this segment.

A passive optical network (PON) uses a point-to-multipoint (P2MP) topology in which a single optical line terminal (OLT) at the exchange serves multiple optical network units (ONUs) at the customer premises through a tree of passive optical splitters, as illustrated in fig. 1.1. No active components are required in the distribution plant, keeping operational costs low.

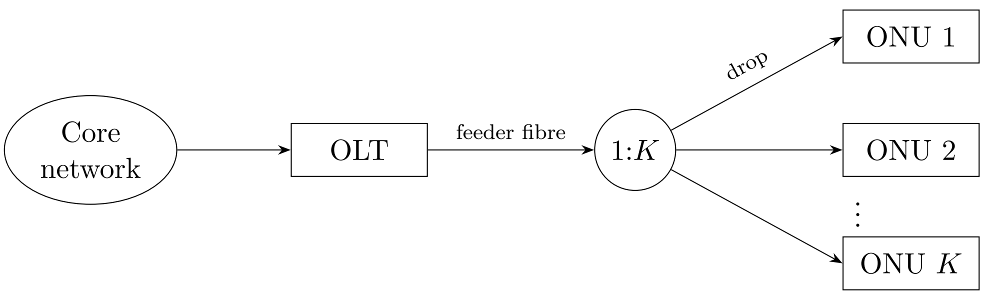


Figure 1.1: Passive optical network: one OLT serves K ONUs via a passive 1:K splitter.

1.2 Coherent PON

Traditional intensity-modulation direct-detection (IM-DD) passive optical networks are approaching their practical limits at data rates approaching 50 Gbps per wavelength. Coherent technology encodes information in the amplitude and phase of the optical signal, allowing modulation formats like QPSK that can carry more bits per symbol. Coherent receivers are also more sensitive, allowing for lower transmit powers. These factors make them a promising alternative to achieve rates of 200 Gbps on an acceptable power budget [5].

1.3 Specification

The Coherent Passive Optical Network (CPON) specification [1] defines a 100 Gbps single-wavelength P2MP system. This will be used as an example to test receiver implementations. This report focuses on the *downstream* direction (OLT to ONU) where receiver optimisations can be shared by all ONUs; the key physical-layer parameters are described in the following sections.

1.3.1 Modulation and Symbol Rate

The downstream signal uses non-differential DP-QPSK at a symbol rate of $R_s = 30.5\,\text{GBd}$. Each polarisation carries 2 bits per symbol by mapping consecutive bit pairs to in-phase and quadrature amplitudes $I, Q \in \{-1, +1\}$, giving QPSK constellation points at $\pm 1 \pm j$. Training and pilot symbols use the same mapping but at amplitude ± 3, placing them outside the data constellation for unambiguous identification.

1.3.2 Downstream Frame Structure

The fundamental framing unit is the *DSP subframe* of 3 712 symbols, organised as 116×32-symbol blocks. Every block begins with a pilot symbol; the first block also carries an 11-symbol training sequence, whose first symbol doubles as pilot index 1.

1.3.3 Key System Tolerances

The standard supports a 35 dB loss budget from OLT to ONU, which it states will allow a link length of 80 km with a 1:16 splitting ratio or 20 km with a 1:512 splitting ratio. To account for impairments in the core network, an SNR of 35 dB at the OLT must be supported. The OLT transmitter laser must be within $\pm 1.5\,\text{GHz}$ of its DWDM channel centre, and the ONU must acquire signals within $\pm 1.8\,\text{GHz}$ of the grid. In the worst case, the combined transmitter and LO offsets yield a carrier frequency offset (CFO) of up to $\approx 3\,\text{GHz}$ that the receiver DSP must correct. Both OLT and ONU lasers must have a linewidth of at most 1 MHz.

The specification requires a pre-FEC bit error ratio of $\leq 2 \times 10^{-2}$, which the soft-decision LDPC FEC reduces to a post-FEC BER of $\leq 10^{-12}$.

Chapter 2

Network and Receiver Models

2.1 Simulation Framework

All DSP algorithms in this report are designed, characterised, and compared through software simulation in MATLAB. The simulated receiver processes DP-QPSK symbols generated at the CPON downstream parameters described in chapter 1.

2.1.1 Fixed-Point Arithmetic

The receiver is evaluated with fixed-point arithmetic, as used in ASIC designs. This captures the effect of quantisation error on each implementation's performance, and underpins a bit-accurate energy model that goes beyond algorithmic complexity. All fixed-point modelling uses the MATLAB Fixed-Point Designer toolkit.

Quantisation Error With b fractional bits the quantisation step is $\Delta = 2^{-b}$; round-to-nearest produces an error e modelled as a zero-mean uniform random variable on $[-\Delta/2, \Delta/2]$ with variance

$$\sigma_q^2 = \frac{1}{\Delta}\int_{-\Delta/2}^{\Delta/2} e^2 \, \mathrm{d}e = \frac{\Delta^2}{12} = \frac{2^{-2b}}{12}. \tag{2.1}$$

For a signal taking values over $[-A, A]$, $\Delta = \frac{2A}{2^b}$ and the signal-to-quantisation-noise ratio in dB with signal energy E_s is

$$\mathrm{SQNR} = 10\log_{10}\left(\frac{E_s}{\sigma_q^2}\right) = 6.02b + 10\log_{10}\left(\frac{3E_s}{A^2}\right). \tag{2.2}$$

CORDIC Coherent receivers perform phase estimates and rotations on complex-valued data in several DSP blocks. This is done efficiently in fixed-point hardware with the CORDIC algorithm, which evaluates these operations using only shifts and additions. CORDIC iteratively rotates a vector by the fixed angles $\arctan(2^{-i})$, each implemented as a bit-shift and an add/subtract. Each iteration adds roughly one bit of angular resolution, so n bits of precision require n CORDIC iterations and the cost is taken to be equivalent to an n-bit real multiplication.

2.1.2 System Parameters

The network under investigation assumes an end-to-end OLT-to-ONU loss budget of 35 dB and a back-to-back transmitter SNR of 35 dB, both taken from the CPON downstream specification of chapter 1. The two configurations are 80 km with a 1:16 splitting ratio and 20 km with a 1:512 splitting ratio. The fibre and signal parameters assumed for the network are listed in table 2.1. Nyquist pulse-shaping is assumed throughout.

Table 2.1: System parameters.

Parameter	Value
Central wavelength (λ)	1550 nm
Channel bandwidth (B)	30.5 GHz
Chromatic dispersion coefficient (D)	17 ps/(nm·km)
PMD coefficient (D_{DGD})	0.1 ps/$\sqrt{\text{km}}$

2.1.3 Channel Impairments

Between the transmitter and receiver, the ideal symbol stream is degraded by the optical channel. Each impairment is applied as a separate stage so that individual DSP blocks can be characterised in isolation or in combination. The models used [8] are described below.

Additive White Gaussian Noise AWGN models the combined effect of shot noise and thermal noise. In simulation, the power of the incoming signal is measured and zero-mean Gaussian noise is added independently to each polarisation at a level set by the target SNR.

Chromatic Dispersion Chromatic dispersion (CD) arises because the group velocity of light in fibre depends on wavelength, so the frequency components of a pulse accumulate a frequency-dependent delay that spreads it over many symbol periods. It is an all-pass effect with a quadratic phase response. It is simulated by transforming the signal to the frequency domain with an FFT, applying the transfer function

$$H(\omega) = \exp\left(j\,\frac{D\lambda^2}{4\pi c}\,L\,\omega^2\right), \tag{2.3}$$

where D is the dispersion coefficient, λ the central wavelength, L the fibre length, and c the speed of light, then transforming back with an IFFT.

Polarisation-Mode Dispersion Polarisation-mode dispersion (PMD) results from random birefringence in the fibre, which causes the two polarisation modes to travel at slightly different speeds and couple into one another along the link. It is simulated as a concatenation of N_{pmd} birefringent sections, each applying a differential group delay of $\pm\tau/2$ to its two principal states in the frequency domain, with a random unitary rotation (the singular vectors of a complex Gaussian matrix) modelling the mode coupling between sections. The mean DGD scales as $\tau \propto D_{\mathrm{PMD}}\sqrt{L}$, with D_{PMD} the fibre PMD coefficient.

Carrier Frequency Offset A carrier frequency offset (CFO) occurs because the transmitter and local-oscillator lasers are free-running and never exactly aligned in centre frequency, as discussed in section 1.3.3. The resulting beat appears as a linear phase ramp on the received samples. It is simulated by multiplying the signal by $\exp(j\,2\pi\,\Delta f\,T\,k)$, where Δf is the frequency offset, T the sample period, and k the sample index.

Phase Noise Phase noise originates from the finite linewidth of the transmitter and LO lasers, whose phases follow a random walk. It is modelled as a Wiener process: independent Gaussian per-symbol phase increments with variance $\sigma^2 = 2\pi\,\Delta\nu/R_s$, where $\Delta\nu$ is the combined laser linewidth, are accumulated and applied as a multiplicative phase $\exp(j\phi[k])$.

2.2 Energy Estimation

A DSP technique's energy efficiency cannot be judged from the receiver alone: a simpler implementation typically requires a higher SNR, which the transmitter must compensate for with extra launch power. The receiver contribution is modelled from fixed-point operation counts, and the transmitter contribution from the minimum SNR needed to meet the pre-FEC BER threshold in the CPON standard [1]. The two are combined to give a per-bit system energy.

2.3 Receiver Energy

The energy consumption of the receiver can be estimated from the number of real multiplications N_M and real additions N_A it performs to process each symbol. The true cost of these operations depends on the hardware implementation. The reference values shown in table 2.2 provide a useful estimate to compare DSP algorithms but a more accurate estimate would require power simulations for an ASIC implementation.

Table 2.2: Energy cost of arithmetic operations for different bit widths in 45 nm CMOS [6].

n (bits)	Add (pJ)	Multiply (pJ)
8	0.03	0.2
32	0.1	3.1

2.3.1 Dependence on Word Length

A ripple-carry adder performs n bit-wise additions on words of length n, so its energy scales linearly with n. A shift-and-add multiplier accumulates n partial products of length n and so scales as n^2. Modern adders and multipliers (e.g. carry-lookahead, Booth/Wallace) reduce these costs by a constant factor or to $n \log n$, so the scalings used here are upper bounds; comparisons between algorithms remain valid because all are penalised equally. The proportionality constants are calibrated by a least-squares fit through the origin to the two data points of table 2.2, giving

$$E_A(n) = 3.16\,n \text{ fJ}, \qquad E_M(n) = 3.03\,n^2 \text{ fJ}. \tag{2.4}$$

Energy consumption for arithmetic operations in modern process nodes is not widely reported, and Dennard scaling (the rule that transistor power density remained constant as they were scaled down) [4] can no longer be applied to extrapolate from the values in table 2.2. As an approximation, the $\approx 75\%$ reduction in total power dissipation at constant frequency reported for an Intel core moving from 45 nm to 14 nm CMOS [9] is taken as a proxy for the per-operation energy scaling. This conflates contributions from leakage, clocks, caches and I/O with the arithmetic energy of interest, so the resulting numbers should be read as an estimate rather than a calibrated figure. Applying this reduction to the 45 nm fit gives

$$E_A(n) = 0.79\,n \text{ fJ}, \qquad E_M(n) = 0.76\,n^2 \text{ fJ}. \tag{2.5}$$

2.3.2 Amortisation

Some DSP algorithms perform computations that are reused over multiple symbols. For example, carrier frequency offset (CFO) varies much slower than the symbol rate so can be estimated less frequently. For a computation that requires N_M multiplications and N_A additions and is used by S symbols, the operations per symbol would be N_M/S and N_A/S.

2.3.3 Total Energy and Power

Many DSP blocks operate on complex-valued data; each complex multiplication is counted as 4 real multiplications and 2 real additions, and each complex addition as 2 real additions. The energy per symbol for the entire receiver is then

$$E_{\text{rx},s} = \sum_{k=1}^{L} 2\left[N_{A,k}E_A(n_k) + N_{M,k}E_M(n_k)\right], \tag{2.6}$$

where the receiver is decomposed into L blocks that use different word lengths n_k according to their required precision, and $N_{A,k}$, $N_{M,k}$ are the addition and multiplication counts per symbol. The factor of 2 is for both polarisations. Blocks operating on the oversampled signal at rate $f > R_s$ have their per-sample operation counts scaled by the oversampling ratio f/R_s. The receiver power is then

$$P_{\text{rx}} = R_s E_{\text{rx},s}, \tag{2.7}$$

and converting to energy per bit allows fair comparison across modulation schemes and symbol rates,

$$E_{\text{rx}} = \frac{E_{\text{rx},s}}{\log_2(M)}, \tag{2.8}$$

where M is the modulation order.

2.4 Transmitter Energy

The downstream link considered here consists of a single OLT transmitting through a 1:K passive splitter to K ONUs, as shown in fig. 1.1. The OLT power is divided equally between branches, so each ONU receives a $1/K$ share.

2.4.1 Minimum Required SNR

The CPON standard specifies a maximum pre-FEC bit-error rate (BER) of 10^{-2} that must not be exceeded for the forward error-correction (FEC) scheme to function correctly. Each receiver implementation therefore has a minimum required SNR to meet this FEC threshold. This SNR, together with the noise characteristics of the optical network, determines the minimum transmitter power.

2.4.2 Noise

The relationship between SNR and transmitter power P_{tx} depends on the dominant source of noise in the network. Optical access networks typically operate over short lengths at low power, so amplifiers can be omitted and non-linear noise neglected, leaving shot noise as the dominant impairment. The shot noise power referred to the optical input of a balanced coherent receiver is $2h\nu B_{\mathrm{eff}}$, where the factor of 2 captures the contributions from the signal and LO arms of the balanced detector summed in quadrature, and B_{eff} is the receiver electrical bandwidth, set equal to the symbol rate by a matched bandpass filter after the 1:K splitter in fig. 1.1. The power per polarisation received by each ONU is $P_{\mathrm{tx}}/2\Gamma K$, where Γ is the loss from OLT to ONU. This gives a per-ONU NSR of

$$\mathrm{NSR} = \frac{4\Gamma K h\nu B_{\mathrm{eff}}}{P_{\mathrm{tx}}} + \mathrm{NSR}_0, \tag{2.9}$$

where $\mathrm{NSR} = 1/\mathrm{SNR}$ and NSR_0 is the back-to-back NSR at the OLT. Rearranging gives the transmitted power as

$$P_{\mathrm{tx}} = \frac{4\Gamma K h\nu B_{\mathrm{eff}}}{\mathrm{NSR} - \mathrm{NSR}_0}. \tag{2.10}$$

Assuming Nyquist pulse-shaping and matched bandpass filtering at the receiver ($B_{\mathrm{eff}} = R_s$), the energy per bit transmitted is

$$E_{\mathrm{tx}} = \frac{P_{\mathrm{tx}}}{2KR_s \log_2(M)} = \frac{2\Gamma h\nu}{(\mathrm{NSR} - \mathrm{NSR}_0)\log_2(M)}. \tag{2.11}$$

2.5 Combined Energy

Before calculating a combined energy contribution, the transmitted optical energy in eq. (2.11) must be converted into an energy drawn from the electrical power supply by estimating the efficiencies of the transmitting laser and the modulator. The dual-polarisation modulator (fig. 2.1) consists of two nested-Mach-Zehnder IQ modulators sharing a common laser through a polarisation beam splitter, with their outputs recombined by a polarisation beam combiner. Each IQ modulator has the transfer function [8]

$$\frac{E_{\mathrm{output}}(t)}{E_{\mathrm{input}}(t)} = \frac{1}{2}\cos\left[\frac{v_I(t)\pi}{2V_\pi}\right] + \frac{j}{2}\cos\left[\frac{v_Q(t)\pi}{2V_\pi}\right]. \tag{2.12}$$

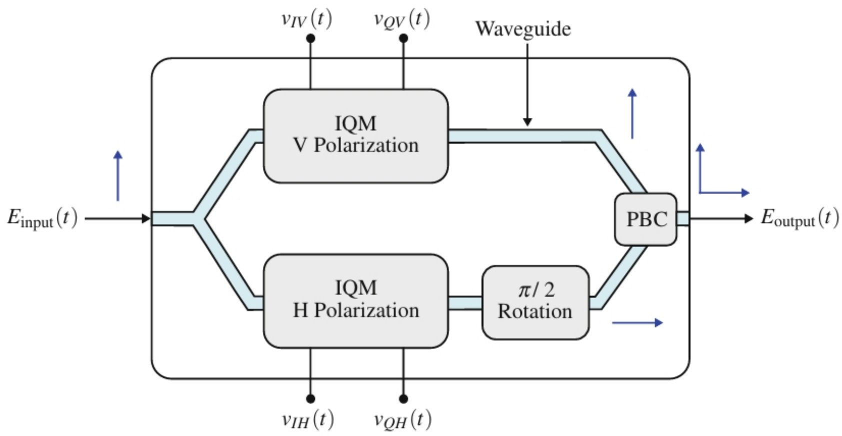


Figure 2.1: Dual-polarisation coherent modulator using MZMs.

With null biasing, QPSK is generated by driving each MZM between transmission minima, $v_I(t), v_Q(t) \in \{0, 2V_\pi\}$, so

$$\frac{E_{\text{output}}(t)}{E_{\text{input}}(t)} = \pm\frac{1}{2} \pm \frac{j}{2} \tag{2.13}$$

and $P_{\text{out}} = P_{\text{in}}/2$, where the loss comes from power coupled to unused ports in the 2×2 couplers. An ideal polarisation beam splitter and combiner introduce no further loss, so the *marginal* efficiency of the dual-polarisation modulator is

$$\eta_{\text{DPM}} = \frac{dP_{\text{tx}}}{dP_{\text{in}}} = 0.5. \tag{2.14}$$

The overall efficiency would also include the power dissipated in the drive electronics, but this is independent of transmit power and so cancels in a comparison of receiver implementations. Combining with the marginal laser efficiency $\eta_{\text{laser}} = dP_{\text{opt}}/dP_{\text{elec}}$ gives a marginal wall-plug efficiency

$$\eta_{\text{WPE}} = \eta_{\text{laser}}\eta_{\text{DPM}} = 0.5\eta_{\text{laser}}. \tag{2.15}$$

η_{laser} will depend on the laser assembly and includes the increase in electrical power to supply the transmit laser and cooling apparatus. The combined energy contribution can now be estimated as

$$E_{\text{sys}} = \frac{1}{\eta_{\text{WPE}}}E_{\text{tx}} + E_{\text{rx}}. \tag{2.16}$$

This does not give the total energy consumed by the optical network, but a difference in E_{sys} shows the energy savings across receiver implementations.

Chapter 3

Equalisation

3.1 Introduction

3.2 Static Equalisation

3.2.1 Equalisation Procedure

Chromatic dispersion is a static, all-pass impairment and is removed by an equaliser that applies the inverse of the channel response. From the channel model of chapter 2, the compensating transfer function is

$$H_{\mathrm{eq}}(\omega) = \exp\left(-j\,\frac{D\lambda^2}{4\pi c}\,L\,\omega^2\right), \tag{3.1}$$

whose inverse Fourier transform gives the continuous-time impulse response

$$h(t) = \sqrt{\frac{c}{jD\lambda^2 L}}\,\exp\left(j\,\frac{\pi c\,t^2}{D\lambda^2 L}\right). \tag{3.2}$$

In the time domain the equaliser is a single FIR filter per polarisation whose taps are eq. (3.2) sampled at the receiver period T and truncated to N_{CD} coefficients. Alternatively, eq. (3.1) can be applied directly in the frequency domain. As the data stream is continuous, the filter is applied block-by-block by overlap-save: each block of N samples is transformed with an FFT, multiplied by the precomputed samples of H_{eq}, and inverse transformed, discarding the first $N_{\mathrm{CD}} - 1$ samples of every block to remove the circular-convolution wrap-around. This leaves $N - N_{\mathrm{CD}} + 1$ valid output samples per block.

3.2.2 Filter Length

The equaliser must span the channel memory introduced by dispersion. The pulse spread across the signal bandwidth is $\Delta\tau = D\,\Delta\lambda\,L$, so the number of taps required is

$$N_{\mathrm{CD}} = \frac{\Delta\tau}{T} + 1, \tag{3.3}$$

where T is the receiver sampling period. With an oversampling rate of 2, table 3.1 lists the taps needed for the two networks of chapter 2.

Table 3.1: Taps to compensate chromatic dispersion for both network models.

L (km)	$\Delta\tau$ (ps)	N_{CD}
20	83	6
80	332	21

3.2.3 Computational Complexity

A radix-2 FFT of size N decomposes into $\log_2 N$ stages, each containing $N/2$ butterflies. Each butterfly applies one twiddle-factor complex multiplication followed by a complex addition and a complex subtraction. With 4 real multiplications and 2 real additions per complex multiplication, and 2 real additions per complex addition, a butterfly costs 4 real multiplications and $2 + 2 \times 2 = 6$ real additions. The full transform therefore requires

$$N_M^{\text{FFT}} = 2N \log_2 N, \qquad N_A^{\text{FFT}} = 3N \log_2 N. \tag{3.4}$$

Each overlap-save block needs a forward FFT, an inverse FFT of equal cost, and N complex multiplications by the precomputed filter spectrum (the filter FFT is computed once offline as the channel is static). This is $2N_M^{\text{FFT}} + 4N$ real multiplications and $2N_A^{\text{FFT}} + 2N$ real additions per block, shared over the $N - N_{\text{CD}} + 1$ valid outputs. Scaling by the oversampling ratio $S = f/R_s$ to express the cost per symbol gives

$$\begin{aligned} N_{M,\text{OS}} &= S\,\frac{4N\log_2 N + 4N}{N - N_{\text{CD}} + 1}, \\ N_{A,\text{OS}} &= S\,\frac{6N\log_2 N + 2N}{N - N_{\text{CD}} + 1}. \end{aligned} \tag{3.5}$$

In the time domain, direct convolution with the N_{CD}-tap complex filter costs N_{CD} complex multiplications and $N_{\text{CD}} - 1$ complex additions per output sample. Converting to real operations and scaling by S, the per-symbol cost is

$$N_{M,\text{TD}} = 4SN_{\text{CD}}, \qquad N_{A,\text{TD}} = S\,(4N_{\text{CD}} - 2)\,. \tag{3.6}$$

The FFT can be simplified by rounding all twiddle-factors to a power of 2. This eliminates all real multiplications in performing the FFT/IFFT, replacing them with bit-shifts of negligible cost.

Operation	Time-domain	Overlap-save	Overlap-save (2^k twiddle)
Real multiplications	$4SN_{\text{CD}}$	$S\dfrac{4N\log_2 N + 4N}{N - N_{\text{CD}} + 1}$	$S\dfrac{4N}{N - N_{\text{CD}} + 1}$
Real additions	$S\,(4N_{\text{CD}} - 2)$	$S\dfrac{6N\log_2 N + 2N}{N - N_{\text{CD}} + 1}$	$S\dfrac{6N\log_2 N + 2N}{N - N_{\text{CD}} + 1}$

Table 3.2: Per-symbol computational cost of the time-domain and overlap-save chromatic dispersion equalisers, including overlap-save with twiddle factors rounded to a power of 2.

Evaluating these expressions with the N_{CD} of table 3.1 gives the per-symbol operation counts for each network in table 3.3.

	Time-domain		Overlap-save		Overlap-save (2^k twiddle)	
L (km)	RM	RA	RM	RA	RM	RA
20						
80						

Table 3.3: Real multiplications (RM) and real additions (RA) per symbol for each chromatic dispersion equaliser implementation, evaluated with the N_{CD} and N of table 3.1.

3.3 Adaptive Equalisation

3.3.1 Equalisation Procedure

Tracking PMD is done with a 2x2 MIMO equaliser that employs 4 finite impulse response (FIR) filters to give outputs

$$x_{\mathrm{out}} = \mathbf{h}_{xx}^H \mathbf{x}_{\mathrm{in}} + \mathbf{h}_{xy}^H \mathbf{y}_{\mathrm{in}}, \quad y_{\mathrm{out}} = \mathbf{h}_{yx}^H \mathbf{x}_{\mathrm{in}} + \mathbf{h}_{yy}^H \mathbf{y}_{\mathrm{in}}. \tag{3.7}$$

The filter weights are updated using an error estimate as

$$\begin{aligned} \mathbf{h_{xx}} &\leftarrow \mathbf{h_{xx}} + \mu e_x \mathbf{x}_{\mathrm{in}} x_{\mathrm{out}}^*, \quad & \mathbf{h_{xy}} &\leftarrow \mathbf{h_{xy}} + \mu e_x \mathbf{y}_{\mathrm{in}} x_{\mathrm{out}}^*, \\ \mathbf{h_{yx}} &\leftarrow \mathbf{h_{yx}} + \mu e_y \mathbf{x}_{\mathrm{in}} y_{\mathrm{out}}^*, \quad & \mathbf{h_{yy}} &\leftarrow \mathbf{h_{yy}} + \mu e_y \mathbf{y}_{\mathrm{in}} y_{\mathrm{out}}^*. \end{aligned} \tag{3.8}$$

where the convergence parameter μ is chosen to be a power of 2 for hardware simplicity.The error terms are produced blindly using the constant modulus algorithm (CMA) [**coherent˙receivers**], which exploits the fact that an ideal QPSK constellation lies on a circle of fixed radius. CMA minimises the cost function $J = \mathbb{E}\left[(|x_{\mathrm{out}}|^2 - R)^2\right]$, where the dispersion constant $R = \mathbb{E}[|s|^4]/\mathbb{E}[|s|^2]$ is set by the transmitted symbols s and reduces to a constant for QPSK. Taking the stochastic gradient of J yields the error estimates

$$e_x = \left(R - |x_{\mathrm{out}}|^2\right), \quad e_y = \left(R - |y_{\mathrm{out}}|^2\right). \tag{3.9}$$

The update step from eq. (3.8) can be simplified by only considering the signs of the error and output terms with no performance penalty [**Cardenas:14**]. The output is a complex number so a complex signum function is defined as

$$csgn(z) = sgn(Re(z)) + jsgn(Im(z)). \tag{3.10}$$

3.3.2 Filter Length

There must be sufficient taps in the filter to cover the channel memory introduced by PMD. The mean differential group delay (DGD) $\langle \Delta\tau \rangle = D\sqrt{L}$ where D is the DGD parameter and L is the link length. The instantaneous DGD is modelled by a Maxwellian distribution [**Mantzoukis:11**]

$$\mathbb{P}[\Delta\tau > x] \approx \frac{4}{\pi} \frac{x}{\langle \Delta\tau \rangle} \exp\left(-\frac{4x^2}{\pi \langle \Delta\tau \rangle^2}\right). \tag{3.11}$$

The number of taps can be found for a given outage probability from eq. (3.11) as

$$N = \frac{x}{T} + 1 \tag{3.12}$$

where T is the sampling period of the receiver. For an access network where $L \leq 80$ km, and a typical DGD parameter $D = 0.1$ ps/$\sqrt{\text{km}}$ then $\langle \Delta\tau \rangle \leq 0.9$ ps. With an oversampling rate of 2, the sampling period $T = 16.4$ ps. Given that an outage probability of 10^{-12} only requires $x = 4.81\langle \Delta\tau \rangle = 4.3$ ps, few taps will be needed for reliable performance.

3.3.3 Computational Complexity

The per-sample cost of the weight updates is compared below for the direct and sign-sign implementations with N taps.

Operation	Direct	Sign-Sign
Real multiplications	$16N + 4$	0
Real additions	$16N$	$16N$

Table 3.4: Per-symbol computational cost of the direct and sign-sign weight updates.

The adaptive equaliser must be parallelised to support a 30.5 GBd symbol rate. The input samples are split into P lanes, with the input vector of lane p

$$\mathbf{x}_p = [\mathbf{x}[mP + p], \mathbf{x}[mP + p - 1], \ldots, \mathbf{x}[mP + p - (N - 1)]] \,. \tag{3.13}$$

The update step for the shared weights is given by

$$\Delta\mathbf{h} = \mu \sum_{p=0}^{P-1} e[p]\mathbf{x}_p x_{\text{out}}[p]. \tag{3.14}$$

A natural choice for blocks of 32 symbols is $P = 32$, allowing the pilot symbols with a larger modulus to be processed separately. With an oversampling rate of 2, each lane would need to operate with a sampling frequency of $2 \cdot 30.5/32 = 1.9$ GHz which is reasonable for modern CMOS. This scales the number of operations in table 3.4 by P.

3.3.4 Fixed Point Precision

The convergence parameter μ in eq. (3.8) is typically $\approx 10^{-3}$ which would require at least 10 bits of precision to represent. Computing the update term to this precison adds unnecessary computational cost, so it should instead be computed at lower precision and bit-shifted (equivalent to setting μ to a power of 2) to the least significant bits of the weights before updating. This can lead to the most significant bits being redundant. These bits waste energy when the weights change sign and when forming the outputs in eq. (3.7) if the weights are negative. Jiang et al. [**Jiang;26**] propose removing these redundant bits with an optimisation over the precision of each weight and demonstrate a decrease in energy consumption with a slight performance loss. The decrease in energy consumption can be estimated directly from the frequency of sign changes in the weights and the proportion of multiplications in eq. (3.7) that use negative weights. For d redundant bits, a sign change is equivalent to the cost of a d-bit addition and the additional cost in forming the outputs will be a d-bit multiplication.

3.4 Methodology

Chapter 4

Carrier Recovery

4.1 Introduction

A coherent receiver recovers the transmitted symbols by mixing the incoming optical signal with a local oscillator (LO) laser, allowing the amplitude and phase of each symbol to be extracted directly. However, the transmitter and LO lasers will in general have different centre frequencies and independent phase trajectories, producing a carrier frequency offset (CFO) and accumulated phase noise on the received symbols. Carrier recovery is the DSP stage responsible for estimating and removing these impairments before symbol decisions are made.

Carrier recovery is divided into two sequential stages. Frequency recovery performs a coarse estimate of the CFO and corrects it so that only slow residual phase variations remain. Phase recovery then tracks and removes these residual phase errors on a block-by-block basis. The algorithm chosen for each stage determines both the minimum achievable SNR and the receiver DSP energy, making algorithm selection central to the system-level energy comparison of chapter 2.

This chapter describes the Rife and Boorstyn (Rife & Boorstyn) and Differential Kay (differential phase and Kay) frequency recovery algorithms and the Viterbi & Viterbi (V&V) and pilots-only phase recovery algorithms, analysing their estimation accuracy and computational complexity. The optimal fixed-point word length for each combination is then found, and the resulting system energies are compared across shot-noise and amplified link scenarios.

4.2 Frequency Recovery

4.2.1 Isolating CFO

All frequency recovery algorithms first require that the phase of the data be removed before estimation. For QPSK symbols $c[i] = e^{j\frac{\pi}{4}+m[i]\frac{\pi}{2}}$, this can be done by raising each received symbol $x[i] = c[i]e^{j(2\pi f_d i+\theta[i])} + n[i]$ to the fourth power to give

$$z[i] = x^4[i] = e^{j(8\pi f_d i+4\theta[i])} + n'[i] \tag{4.1}$$

where θ is phase noise, f_d is the CFO, and n is additive noise. This allows all data symbols to be used in CFO estimation, but amplifies phase and additive noise, which degrades the estimation quality. The fourth power operation also requires three complex multiplications. An alternative is to use pilot symbols where the data phase is known beforehand, giving

$$z[i] = x[i]c^*[i] = e^{j(2\pi f_d i+\theta[i])} + n[i], \tag{4.2}$$

which is more robust to noise and less computationally expensive. Some algorithms only require $\arg(z[i])$, which is efficiently found using CORDIC as

$$\arg(z[i]) = \{4\arg(x[i])\}|_{-\pi}^{\pi} \tag{4.3}$$

for blind operation or

$$\arg(z[i]) = \{\arg(x[i]) - \arg(c[i])\}|_{-\pi}^{\pi} \tag{4.4}$$

for data-aided operation, where the wrap operation can be done with no additional cost by aligning the wrap range to the overflow range of the fixed-point value used to store the phase.

4.2.2 Rife and Boorstyn

The value of the CFO will appear as the strongest peak in the frequency spectrum of $z[i], z[i+1], \ldots, z[i+L]$ where L is the observation length. The Rife and Boorstyn algorithm first performs an FFT on the observed sequence and finds the index k corresponding to the strongest peak. This coarse estimation is refined with a parabolic interpolation on the complex FFT values X_{k-1}, X_k, X_{k+1} to give [7]

$$\delta = -\Re\left[\frac{X_{k+1} - X_{k-1}}{2X_k - X_{k-1} - X_{k+1}}\right]. \tag{4.5}$$

The updated index $k' = k + \delta$ is then scaled by the sampling rate to find the true CFO. The resolution in k is $\epsilon(k) = 1/2N_{FFT}$. The CPON specification provides only $L = 11$ training symbols, so data-aided operation without zero-padding gives $\epsilon(\hat{f}_d) \approx 0.05\,R_s$ — too coarse for the downstream phase recovery to correct. Zero-padding the training sequence to a larger N_{FFT} improves the resolution without requiring additional training data. The normalised mean squared-error (MSE) in the estimated CFO is shown in fig. 4.1 for data-aided operation with 11 training symbols. A CFO estimate is computed for both polarizations and averaged to reduce the MSE. The modified Cramer-Rao bound (MCRB) provides a lower bound on the achievable MSE and is given by [3]

$$\text{MCRB} = \frac{3}{2\pi^2 L^3 \times \text{SNR}}. \tag{4.6}$$

Rife & Boorstyn achieves this bound above a threshold SNR of ≈ 4 dB, where the threshold SNR is defined as the SNR below which the estimator begins to deviate from the MCRB. Below this, the noise has a significant contribution to the FFT values and can lead to an incorrect index being chosen.

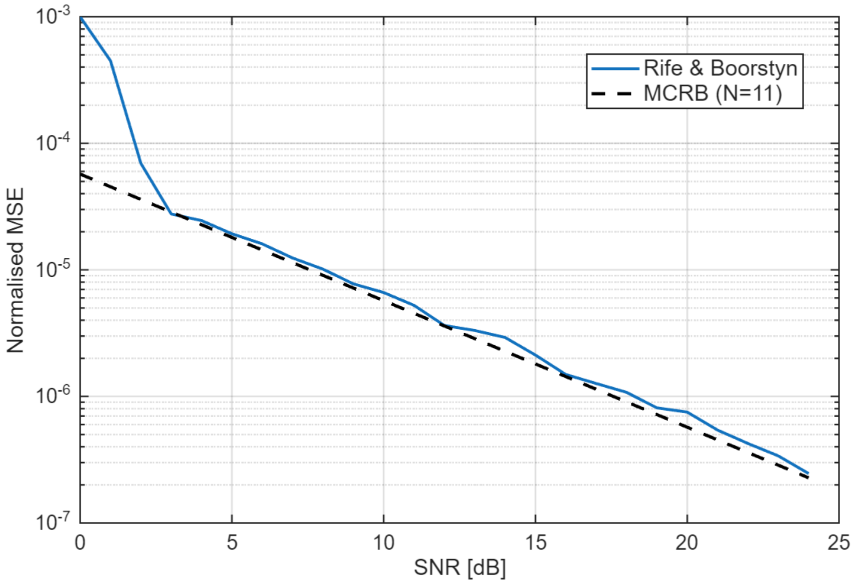


Figure 4.1: Normalised MSE of the Rife & Boorstyn algorithm with FFT size $N_{FFT} = 512$ compared to the MCRB for $L = 11$ training symbols.

Data-aided operation is constrained by the length of the training sequence, but the accuracy of the Rife & Boorstyn algorithm improves with longer observation lengths, as shown in fig. 4.2 for non-data-aided operation. The SNR threshold at which this algorithm begins to deviate from the MCRB is initially higher with data-aided operation due to the amplification of noise, but decreases for longer observation lengths.

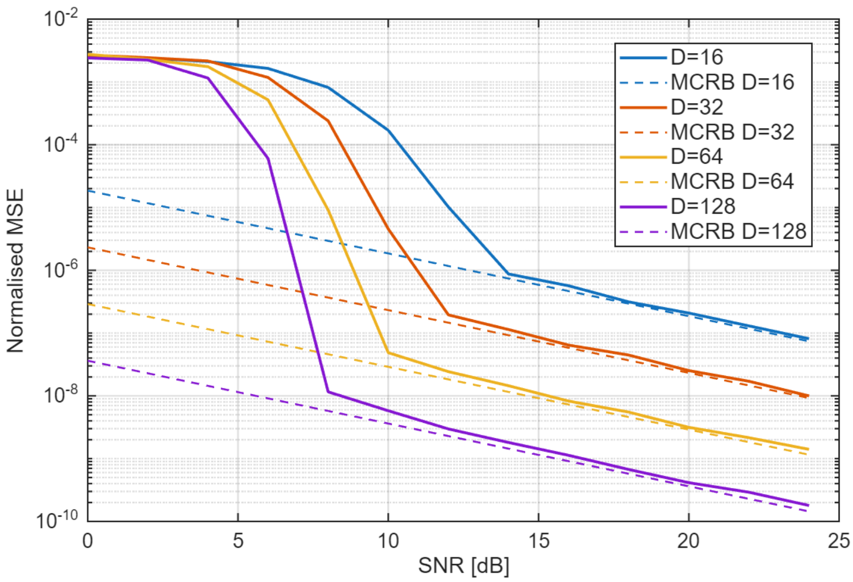


Figure 4.2: Normalised MSE of the Rife & Boorstyn algorithm for non-data-aided operation and different observation lengths.

4.2.3 Differential Phase and Kay Estimator

The Rife & Boorstyn algorithm exhibits good performance even at low SNRs, but is computationally expensive. The FFT requires $\frac{N_{FFT}}{2}\log_2(N_{FFT})$ complex multiplications and $N_{FFT}\log_2(N_{FFT})$ complex additions. The fine search from eq. (4.5) also requires a complex division. A cheaper approach would involve computing the phase difference

$$\arg(z[i]) - \arg(z[i-1]) = 2\pi f_d T_s + \theta[i] - \theta[i-1] + \eta[i] \tag{4.7}$$

over several symbols and averaging to remove the effect of θ. Estimating f_d in this manner is vulnerable to the effect of additive noise on the phase $\eta[i]$, so exhibits poor accuracy. This can be improved with a second pass using the Kay estimator. The Kay estimator uses a least-squares approach, computing the estimated CFO as

$$\hat{f}_d = \frac{R_s}{2\pi}\sum_{k=1}^{L}\frac{6k(L-k)}{L(L^2-1)}\arg(z[k]z^*[k-1]). \tag{4.8}$$

where $\arg(z[k]z^*[k-1])$ computes the phase difference from eq. (4.7) and $\frac{6k(L-k)}{L(L^2-1)}$ is a weighting term that prioritises the contribution of phase differences in the middle of the observed sequence. This estimator is able to achieve the MCRB, but the least-squares approach considers phase differences of non-adjacent symbols, making it vulnerable to phase wrapping that distorts the estimate when the CFO is high enough. When the total phase drift $\Delta\phi = 2\pi f_d L T_s$ across the observation window exceeds 2π, a wrap shifts the estimated CFO by $\pm 1/(LT_s)$. A first-pass coarse CFO correction must therefore be applied before the Kay estimator to keep the residual CFO within the unambiguous range. The performance of the differential phase + Kay estimator for data-aided operation is shown in fig. 4.3. The MCRB is achieved above a threshold SNR of ≈ 7 dB. This is higher than for Rife & Boorstyn, but at a reduced computational cost. Unlike the Rife & Boorstyn algorithm, the threshold SNR of the differential phase and Kay estimator does not depend on observation length, as shown in fig. 4.4.

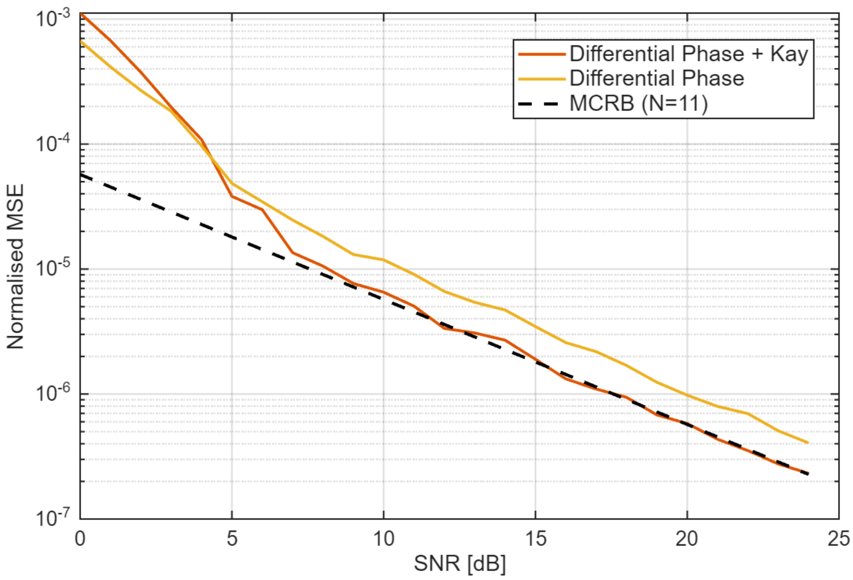


Figure 4.3: Normalised MSE of frequency estimation with the differential phase and Kay estimator for $L = 11$ training symbols.

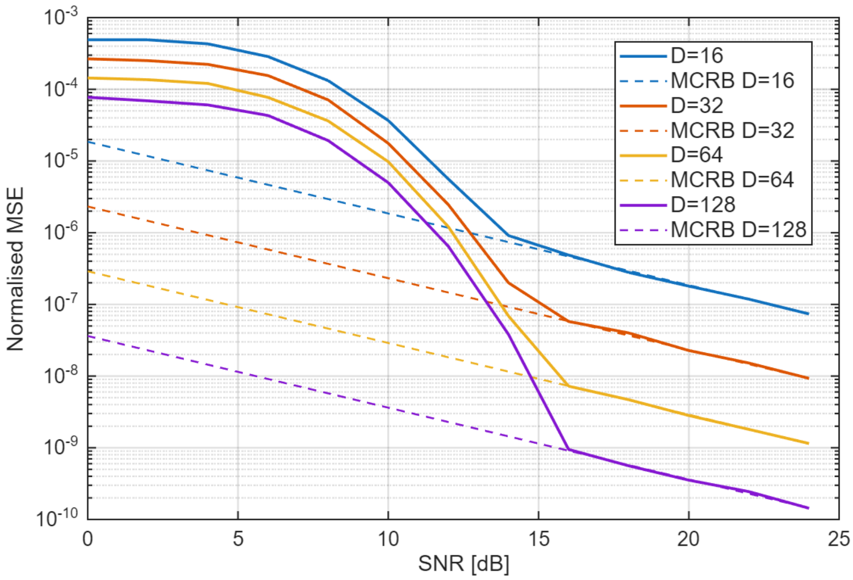


Figure 4.4: Normalised MSE of the differential phase and Kay estimator with blind operation for different observation lengths.

4.2.4 Computational Complexity

Here, the computational complexity of both frequency recovery algorithms is compared in terms of real multiplications and real additions. CORDIC is used to find phases and is assumed to cost the same as a real multiplication. Real division is also assumed to cost the same as multiplication since it is only performed once (this would be a suitable approximation if look-up tables are used to find reciprocals at some cost to accuracy). All analysis applies to a single polarization. If averaging is done over both polarizations, then all operations are doubled.

Rife & Boorstyn The Rife & Boorstyn algorithm requires an FFT followed by N_{FFT} comparisons of the magnitudes in the coarse search. Many of the FFT entries are 0 due to padding, which reduces the total number of operations needed. The interpolation requires the real part of a complex division

$$\Re\left[\frac{a+bj}{c+dj}\right] = \frac{ac+bd}{c^2+d^2}. \tag{4.9}$$

The total complexity is summarised in table 4.1, with the complexity for blind operation marked with (*).

Operation	RM	RA
Forming $z[i],\dots,z[i+L]$	$4L$, $12L^*$	$3L$, $9L^*$
FFT	$2L\log_2(\frac{N_{FFT}}{L}) + 2N_{FFT}\log_2(L)$	$3L\log_2(\frac{N_{FFT}}{L}) + 3N_{FFT}\log_2(L)$
Search	$2N_{FFT}$	$2N_{FFT}$
Interpolation	5	4

Table 4.1: Computational complexity of estimating CFO with the Rife & Boorstyn algorithm.

Differential phase and Kay The differential phase and Kay estimator has lower complexity, only requiring CORDIC, real additions, and real multiplications. The fixed multiplications by the weights of the Kay estimator can be optimised into a low-energy shift-add circuit, but this is ignored for simplicity. The first-pass CFO correction involves forming the phase corrections of each training symbol as $\phi[i] = \hat{f}_d i$, then applying them with CORDIC.

Operation	RM	RA
Forming $\arg(z[i]), \ldots, \arg(z[i+L])$	$2L$	L
Differential Phase	1	$L-1$
First Pass Correction	$2L$	0
Forming $\arg(z[i]), \ldots, \arg(z[i+L])$	$2L$	L
Kay	L	$L-1$

Table 4.2: Computational complexity of estimating CFO with the differential phase and Kay estimator.

4.2.5 Optimising Bit Width

These frequency recovery algorithms compute an estimate of the normalised CFO f_d/R_s. For n bits of precision, the smallest normalised CFO that can be estimated is 2^{-n} for an estimation range $0 \leq |f_d/R_s| \leq 1$. In reality, a much smaller range of CFOs needs to be estimated. If the maximum permitted value of $|f_d/R_s| = 2^{-d}$, d bits of precision that would otherwise be redundant can be utilised by scaling each estimate by 2^d during calculation, then inserting d bits in the final answer to recover the true normalised CFO. This improves the resolution to $2^{-(n+d)}$ without any increase in energy consumption.

4.2.6 Maximum CFO

The theoretical limit on the maximum CFO that can be corrected is given by Nyquist as $0.5R_s$. From eq. (4.1), we see that blind operation reduces this limit by a factor of 4. Both the Rife & Boorstyn and differential phase and Kay estimators accurately estimate the CFO up to the Nyquist limit, as seen in fig. 4.5. Blind operation aliases as expected at $f/R_s = 0.125$, making it viable for CFOs up to 3.8 GHz with a symbol rate of 30.5 GHz.

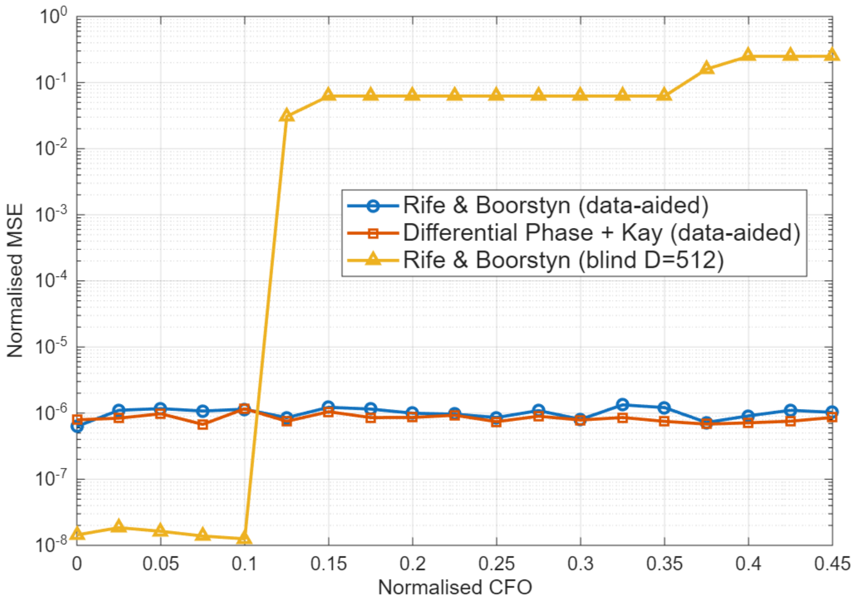


Figure 4.5: Normalised MSE in CFO estimate (MSE^2/R_s^2) vs Normalised CFO (f/R_s)

4.3 Phase Recovery

4.3.1 Feed-Forward vs PLL Algorithms

One of the first techniques investigated for phase recovery was a digital phase-locked loop (PLL) to incrementally correct the phase of each symbol with feedback. This dramatically simplifies the phase estimation and correction due to the use of small-angle approximations. The CPON specification sets the symbol rate at 30.5 GBd. With an oversampling rate of 8/7, the total samples per second would be 35 GSa/s. This is beyond the achievable clock speed of CMOS, so symbols must be processed in blocks of e.g., 32 symbols with a clock speed of 1.1 GHz. This approach hinders the effectiveness of a PLL, as updates can only be made every 32 symbols. For this reason, feed-forward algorithms that can operate on blocks of data were used instead.

4.3.2 Viterbi & Viterbi

The Viterbi & Viterbi algorithm is well suited to QPSK and removes data phase with the fourth power operation of eq. (4.1). It then applies a maximum-likelihood filter $w_{ML}[i]$ to estimate the phase error due to Wiener phase noise from the combined laser linewidth and any residual CFO as

$$\hat{\theta}_{ML}[i] = \frac{1}{4} \arg\left(\sum_{k=1}^{N} w_{ML}^*[k]z[k]\right) - \frac{\pi}{4}. \tag{4.10}$$

A phase unwrapper is then applied before the phase error is corrected. This is readily adapted to block-based processing. If blocks of 32 symbols are used again, the ML filter can be applied to all symbols in the block and the estimated phase error applied to the entire block. This is only valid if the phase difference between the start and end of the block is small, meaning the laser linewidth cannot be too large and the preceding frequency recovery algorithm must correct most of the CFO. By increasing the time between phase estimates, block processing also increases the chance of cycle-slips in the phase unwrapper. The CPON specification provides 1 pilot symbol in every 32-symbol block that can be used to correct this [2]. The correction estimates the number of $\pi/2$ phase jumps n that occur between the unwrapped phase of a data symbol $\hat{\theta}_{PU}$ and its corresponding pilot, then adjusts the unwrapped estimate as $\hat{\theta}'_{PU} = \hat{\theta}_{PU} - n\frac{\pi}{2}$.

4.3.3 Pilots-Only Correction

While Viterbi & Viterbi is commonly used for QPSK, a simpler approach is to compute the phase error from the pilot symbol using eq. (4.2) and apply this to the whole block. This also relies on a small phase change across the block and is more vulnerable to additive noise. However, QPSK allows for $\pm 45°$ error in the phase estimate before an incorrect decision is made, which makes it robust to the drop in accuracy associated with this approach.

4.3.4 Computational Complexity

Viterbi & Viterbi For a block-based approach, the length of the filter will be the block length. Phases are again found with CORDIC. The cost of the phase estimate for each block is shown in table 4.3.

Operation	RM	RA
Forming $z[i], \dots, z[i+N]$	$12N$	$9N$
Computing $\hat{\theta}_{ML}$	$4N+1$	$3N+2(N-1)+1$

Table 4.3: Computational complexity of block-based Viterbi & Viterbi.

Pilots-Only Only phases are needed, so the entire estimation can be performed with CORDIC. The pilot phase is also known beforehand, so does not contribute to the computational cost. The total cost of the phase estimate is shown in table 4.4.

Operation	RM	RA
Forming $\arg(z[i]) = \hat{\theta}$	1	1

Table 4.4: Computational complexity of Pilots-Only correction.

4.4 Methodology

4.4.1 Phase Recovery Comparison

The two phase recovery algorithms are compared in isolation by simulating an AWGN channel with additive Wiener phase noise, without chromatic dispersion, PMD, or fibre nonlinearity, so that any performance difference is attributable solely to the phase estimation stage. BER is estimated by Monte Carlo simulation over independent realisations for each combination of laser linewidth and SNR, with the phase ambiguity inherent to QPSK resolved by exhaustive search over the four $\pi/2$ rotations before symbol decisions are made.

4.4.2 System Energy Comparison

4.5 Results & Discussion

4.5.1 Phase Recovery Performance

To replace Viterbi & Viterbi with the simplified pilots-only phase correction, it must exhibit comparable performance over the range of linewidths that could be expected during operation of the network. The BER for each phase recovery algorithm as SNR varies is shown in fig. 4.6. Both algorithms have almost identical performance (plots overlapping) and are reliable up to 10 MHz, well beyond the maximum of 1 MHz given in the CPON spec.

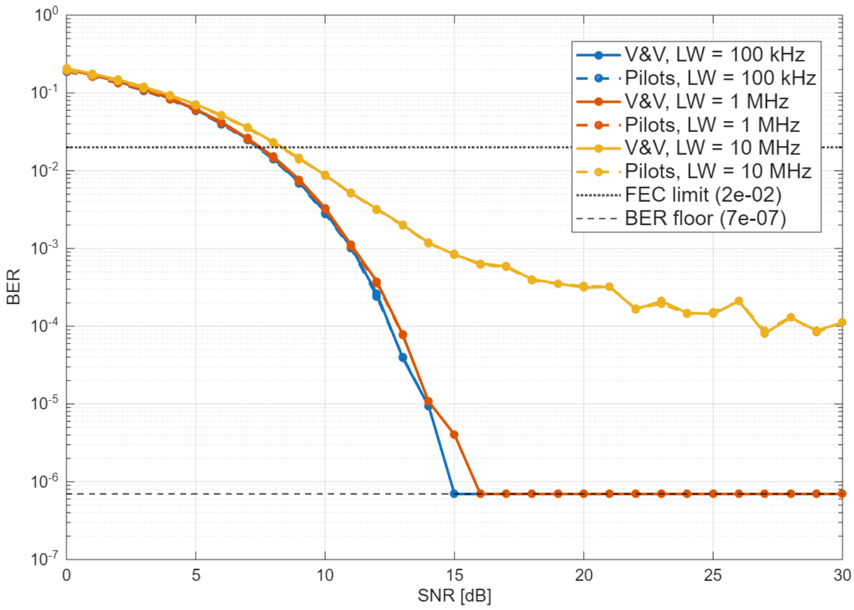
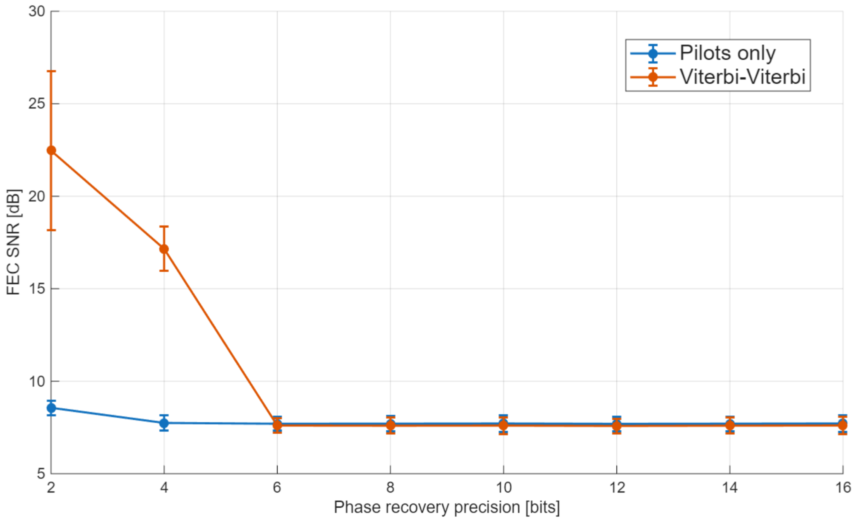
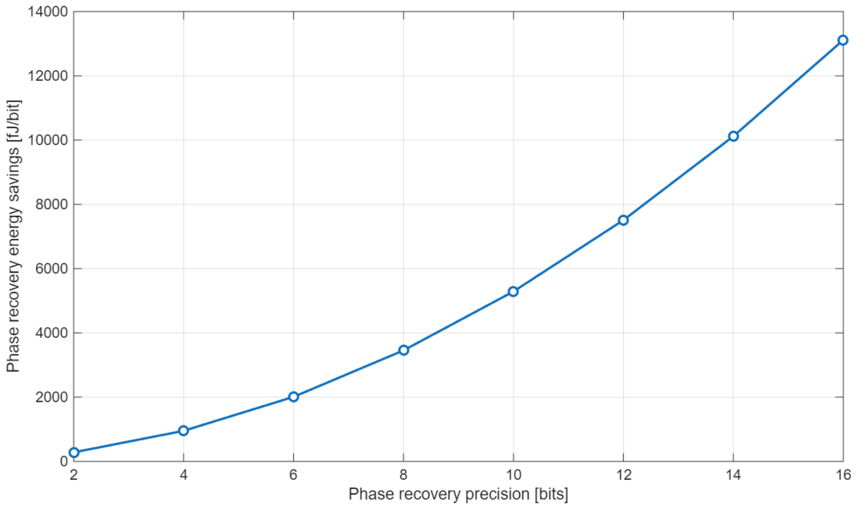


Figure 4.6: Performance comparison of V&V and Pilots-Only phase correction

The choice of fixed-point precision used in phase recovery will balance receiver energy and performance. Figure 4.7a compares the performance of both algorithms and fig. 4.7b shows the energy savings when using pilots-only instead of Viterbi & Viterbi. Both algorithms perform almost identically down to 6 bits, where the FEC SNR of Viterbi & Viterbi rises quickly. The additional operations needed by Viterbi & Viterbi to estimate the phase error mean that quantisation errors compound, degrading perfomance at low precision. The ability of the pilots-only estimate to match or improve on the FEC SNR of Viterbi & Viterbi is likely due to the inherent robustness of QPSK to additive and phase noise; pilots-only will produce a slightly less accurate estimate but as long as it brings the symbol within the correct QPSK quadrant, decoding will be succesful. It is then clear from figs. 4.7a and 4.7b that a receiver can save energy at no performance cost by using the pilots-only estimate with 4 bits of precision. The increase in transmitter energy from the slight increase in FEC SNR at 2 bits of precision may be outweighed by the receiver energy savings, but this will depend on if the carrier recovery block limits the receiver SNR and other system parameters like the wall-plug efficiency of the laser.



(a) FEC SNR.



(b) Energy saved by switching V&V $\rightarrow$ pilots-only.

Figure 4.7: Phase recovery as fixed-point precision varies with 1 MHz laser linewidth: FEC SNR for pilots-only and Viterbi & Viterbi, and the phase recovery energy saved by replacing V&V with pilots-only. Rife & Boorstyn used to remove initial CFO.

4.5.2 Blind vs Data-Aided Operation

The decrease in threshold SNR for longer observation lengths suggests that the performance of the Rife & Boorstyn algorithm will improve with blind observation length D. This is not the case for the differential phase and Kay estimator, where increasing blind observation length has no effect on threshold SNR. The effect of observation length on FEC SNR is shown for both algorithms in fig. 4.8, where blind Rife & Boorstyn shows a clear increase in performance over the blind differential phase and Kay estimator for longer observations and improves on data-aided Rife & Boorstyn after ≈ 50 symbols. The differential phase and Kay estimator in blind operation is thus not viable for frequency recovery, but blind Rife & Boorstyn can provide the best performance at a higher computational cost.

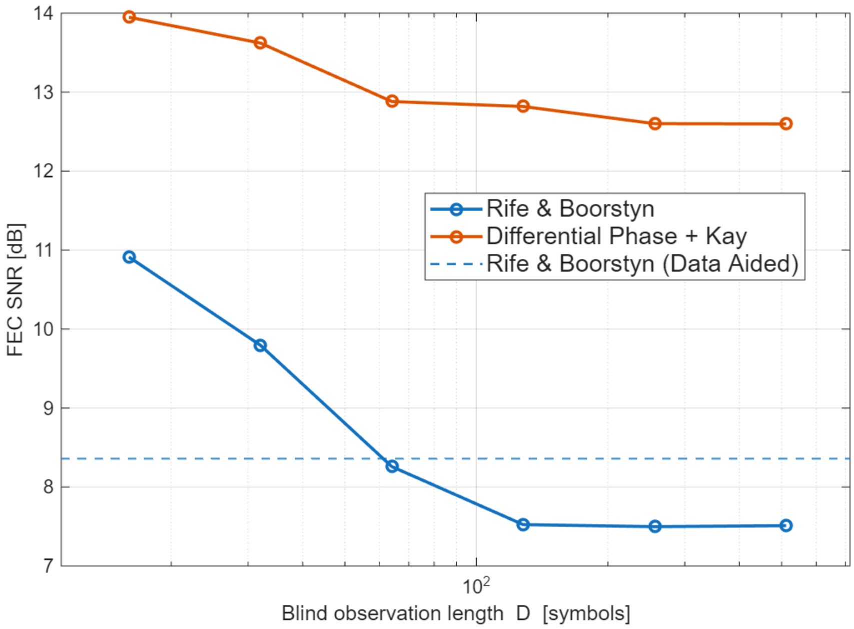


Figure 4.8: FEC SNR vs blind observation length for the Rife & Boorstyn and differential phase and Kay estimators. CFO = 2 GHz, laser linewidth = 1 MHz.

4.6 Conclusions

The robustness to both additive and phase noise of QPSK allows the phase recovery block of a coherent receiver to be dramatically simplified by just using pilot symbols with no noticeable drop in performance, resulting in considerable energy savings. Frequency recovery can also be implemented with simple algorithms to save energy; however, the efficacy of doing so depends on how often the CFO must be estimated. For access networks dominated by shot noise where dispersive effects have been compensated, carrier recovery performance is independent of the length and splitting ratio when constrained to a given loss from OLT to ONU.

Chapter 5

Full DSP Pipeline

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